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1. Word: “Holds”

   Precision value: 1

   The precision for “holds” with respect to the positive class is 1.0, indicating that the word consistently appears in positive reviews, even though it is not an objectively positive adjective.

   In the review, it appears in the sentence *“CARRIE still holds the title for best King adaptation as far as I'm concerned; but SEMATARY is right up there in the Top Five.”*

   In this context, “holds” commends the movie, and it suggests that the film maintains an admired status. The high precision score reflects a pattern where “holds” consistently is used in a context that signals approval.

2. Word: Disturbing

   Precision Value: 0.5

The word “disturbing” has a precision value of 0.5, which means it appears equally in positive and negative reviews. This comes across as surprising since “disturbing” is typically perceived as a negative word.

In this positive review, *“It is somewhat disturbing and kids of course should be shielded from it but this is great dramatic cinema,”* the word “disturbing” describes the film’s content rather than its quality. The reviewer ultimately expresses approval at the end of the sentence, despite acknowledging the unsettling nature of the film.

In contrast, in this negative review, *“In her first novel she wove together a dark and deeply disturbing tale of faith, doubt and redemption,”* the term describes the quality of the movie. Placing more emphasis on words toward the end of the review could be very helpful, as people often summarize their overall opinion in the conclusion. By giving more weight to these final statements the classifier could better capture the reviewer’s true sentiment.

3. After training the model on train.csv and reapplying it to the same dataset, the results showed 10 false negatives and 7 false positives, as indicated in Table 1.

Act/Pred	neg	pos
neg	244	7
pos	10	239

Table 1. The predicted and actual values of the trained train.csv file.

When comparing performance, the model performed better on the training set than on the test set, which indicates overfitting. The model learned patterns specific to the training data that did not generalize well to new data.

Dataset	Accuracy (%)
train.csv (trained and tested on same data)	93%
test.csv (trained on train.csv, tested on new data)	79%

False Negative

"I saw Jack Frost for Â£4:00 at my local store and I thought it looks pretty good for a low budget movie so I bought it and I was right it was good. For starters this film is about a killer snowman so that's something to laugh about and the way it looks was funny compared to the Snowman on the cover. The acting was okay and the lines Jack Frost said had me laughing I only axed you for a smoke and Worlds most pi**ed off snow cone how funny and camp is that? The tale at the start was pretty funny and silly too Jack be nimble Jack be quick Jack gouged eyes with candle sticks . If you're looking for a B-Movie Comedy horror that's full of puns then check Jack Frost out. 10/10"

The review of *Jack Frost* was misclassified as negative despite being positive especially when the reviewer expresses their positive sentiments for the movie through humor and praise. Because Naive Bayes relies primarily on word frequency, the lack of explicitly positive terms such as *"amazing"* or *"wonderful"* likely caused the classifier to

misinterpret the sentiment. I think another reason is another contributing factor could be the use of unigrams, which do not account for context of words.

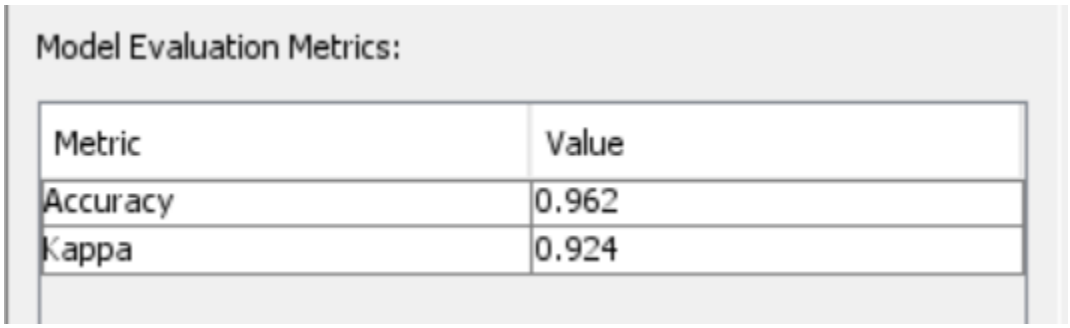
4.

Threshold (t)	Training Accuracy	Test Accuracy
1	0.992	0.780
2	0.986	0.800
4	0.976	0.806
6	0.948	0.802
8	0.930	0.768
10	0.914	0.758

As the threshold (t) increases, the model starts ignoring words that do not appear “t” times in the training data. This helps prevent overfitting because it focuses on words that show up more consistently across reviews. The best performance happened at t = 4, where the test accuracy reached 0.806. This suggests that keeping some, but not all, of the rare words helps the model make better predictions. When the threshold is too low, the model learns from noise; when it is too high, it loses useful information. Overall, t = 4 - t = 5 seems to provide the best balance between detail and accuracy.

5. To improve the accuracy of my Naive Bayes classifier on *test.csv*, I experimented with several different feature representations in LightSIDE. First, I adjusted the feature threshold (t), which determines how frequently a term must appear in the training data to be included. Starting from *t = 1* (all words included), the model performed very well on the training set but poorly on the test set, showing clear signs of overfitting. As I increased the threshold to *t = 4*, the model's test accuracy improved, reaching the best balance between training performance and generalization. However, when I pushed the threshold further to *t = 8* and *t = 10*, performance declined again, suggesting that too many informative words were being excluded.

   I also experimented with different types of features from the configuration pane. Using *unigrams* gave solid baseline results, but adding *bigrams* (two-word combinations) slightly improved test accuracy, as it helped capture context, such as phrases like "not good" or "very funny", that single words alone can miss. The best classifier was 9.6.



Model Evaluation Metrics:

Metric	Value
Accuracy	0.962
Kappa	0.924